

MAY ZIN MG

Data Engineer | Azure Data Platforms | SQL | ETL/ELT | Lakehouse

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Microsoft Certified: SQL AI Developer Associate (DP-800) | Azure Solutions Architect Expert | PMP

PROFESSIONAL SUMMARY

Data Engineer and cloud/data platform specialist with 15+ years across enterprise software, backend integrations, databases, and production systems, including recent hands-on work in Azure data platforms, ETL/ELT, SQL, lakehouse architecture, workflow orchestration, data quality, and production automation. Strong background in Azure Data Factory, Synapse, Data Lake/Blob Storage, SQL Server, Parquet, Python, Power BI, CI/CD, and Docker, supported by deep enterprise integration and reliability experience. Completing an MSc in Business Intelligence.

CORE SKILLS

Data Engineering: Azure Data Factory, ETL/ELT, incremental loads, batch processing, Airflow, dbt, DuckDB, data quality checks, validation, reconciliation

SQL & Databases: SQL Server/T-SQL, stored procedures, query optimisation, PostgreSQL, DynamoDB, Redis, high-volume analytical/reporting workloads

Cloud & Platform: Azure Synapse, Functions, Logic Apps, Container Apps, Key Vault, Managed Identity, App Services, enterprise integration

Lakehouse & Modelling: Synapse, Data Lake/Blob Storage, Parquet, bronze/silver/gold medallion layers, dimensional modelling, analytical datasets, KPI aggregates

Python & Streaming: Python ETL scripting, data processing/automation, Kafka project exposure, FastAPI exposure, NDJSON/Parquet processing, stream-to-batch patterns

DevOps & Reliability: Azure DevOps, GitHub Actions, CI/CD, Docker, monitoring, backup/data movement, retry/recovery, observability, production troubleshooting

PROFESSIONAL EXPERIENCE

AZ Digital Pte Ltd, Singapore | 2015 - Present | 11-year progression

Career evolution from enterprise backend/API delivery into Azure cloud data platform engineering, ETL/ELT, lakehouse architecture, workflow orchestration, DevOps, reporting automation, and AI-ready data foundations.

Cloud & Data Platform Engineer | 2020 - Present

- Designed ADF/Synapse lakehouse pipelines for multi-source sales and logistics data using Parquet and medallion-style layers, supporting more reliable reporting and downstream analytics.
- Built and supported ETL/ELT and data-movement workflows across APIs, SQL Server, Azure Storage, and enterprise operational systems, with validation, retry, recovery-friendly execution, and repeatable monitoring patterns.
- Improved selected dashboard and reporting query responsiveness by an estimated 30-40% through SQL Server stored procedure and high-volume query optimisation.
- Reduced recurring manual support/reporting effort by an estimated 30-40% through scheduled automation using Hangfire, Azure Logic Apps, Azure Functions/scripts, and retry/resend workflows.
- Dockerised and deployed cloud-native data/integration services to Azure Container Apps using Key Vault/Managed Identity and CI/CD via Azure DevOps and GitHub Actions.

Backend & Systems Engineer | 2015 - 2020

- Built and maintained .NET/gRPC/REST integration services connecting finance, logistics, POS, membership, reporting, and operational systems, improving data consistency across workflows.
- Developed SQL-backed reporting and operational workflows across finance, POS, logistics, and membership domains, creating reliable source and integration patterns later extended into cloud data platforms.
- Owned production support for business-critical backend and database systems, including troubleshooting, recovery, deployment, and operational reliability.

Earlier roles: CalendarOne, Singapore (2012-2015, 2008-2011) | National Library Board, Singapore (2011-2012) | Theseus Computer Help Centre, Myanmar (2004-2006)

SELECTED DATA & AI PROJECTS

Token Risk Intelligence Lakehouse (In Progress) | Python, Etherscan API, Polygon, JSONL, data-quality/recovery logging
Building a research-oriented ingestion pipeline for public Polygon token-transfer data. Implemented reproducible UTC/block windows, API pagination and rate-limit handling, immutable raw JSONL capture, and recovery/quality logs, with staged analytical processing designed for traceable risk-signal development.

IoT Wind Farm - Edge to Insights | Python, Kafka, Redis, Airflow, DuckDB, Parquet, Docker, GitHub Actions
Built an end-to-end streaming analytics pipeline where 30 mock turbines emit Kafka telemetry; a Redis-deduplicated consumer writes time-bucketed Bronze NDJSON, while Airflow and DuckDB produce Silver Parquet and Gold KPI outputs. Demonstrates streaming ingestion, idempotency, orchestration, medallion architecture, and CI.

ProMan DataPlat ELT Pipeline | Python, Airflow, dbt, DuckDB, Azurite, Parquet, Power BI
Built a local Azure-style ELT platform using blob-style raw storage, Airflow orchestration, dbt/DuckDB bronze-silver-gold models, data-quality checks, and Power BI-ready exports, demonstrating reproducible transformation and analytics delivery patterns.

Linear B - Agentic Data Platform (In Progress) | Python, DuckDB, Airflow, dbt, LLM Integration
Developing a medallion-style research data platform with ingestion, validation, transformation, analytical outputs, and an LLM-assisted interpretation layer, separating deterministic data processing from AI interpretation and human review.

Talos Agent Lab - Agentic AI Learning Series | Python, Microsoft Agent Framework, Foundry Local, Azure OpenAI, pytest
Built a permission-gated tool-calling agent with automated tests and execution tracing; compared local and cloud model behaviour, investigated hallucinated tool/data cases, refined guardrails, and re-validated grounded behaviour. Demonstrates AI-ready engineering, testing, and observability.

EDUCATION & CERTIFICATIONS

MSc Business Intelligence (In Progress) IU International University of Applied Sciences, Germany 2024 - Present	• Microsoft Certified: SQL AI Developer Associate (DP-800)
MSc Information Systems Nanyang Technological University (NTU), Singapore 2012 - 2014	• Microsoft Certified: Azure Databricks Data Engineer Associate (DP-750) - Exam scheduled Sep 26, 2026
BSc Information Systems & Software Engineering Oxford Brookes University 2006 - 2007	• Azure Solutions Architect Expert • PMP • Google Advanced Data Analytics • Oracle Cloud Infrastructure Data Science Professional
Portfolio: github.com/mayzinmg	